The Link Function Problem in Psychological Interaction Testing

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Abstract

Interaction tests are common in psychology, but an interaction claim is not defined by a product term alone. It also depends on the scale on which additivity is assumed. We reframe this scale problem in the GLM language of outcome family and link function: the family describes the response, whereas the link defines the scale on which no product-term interaction is assumed. We first report a preregistered descriptive review of recent articles from five leading psychology journals, showing that interaction tests are common and that family, link, and scale-of-additivity choices are often left implicit even when outcomes have formal constraints. We develop a link-centered framework, grounded in prior work on scale dependence, outcome-model mismatch, and GLM interaction interpretation, that separates family, link, and target-metric choices. The main focus is the case in which the response family is broadly plausible but the chosen link still defines the wrong no-interaction baseline. Worked examples and simulations focus on common psychological response formats: forced-choice accuracy with a non-zero chance floor, within-family link choices such as logit versus probit, and sum scores as a related measurement case. The simulations show that plausible but inadequate links can turn additive main-effect structures on one scale into statistically significant pseudo-interactions on another. A diagnostic simulation shows that residual checks, link checks, and AIC comparisons may help, but they do not by themselves decide which scale is right for the scientific interaction claim. We argue that interaction claims are often underspecified unless researchers state, justify, and probe the scale on which no interaction is assumed.

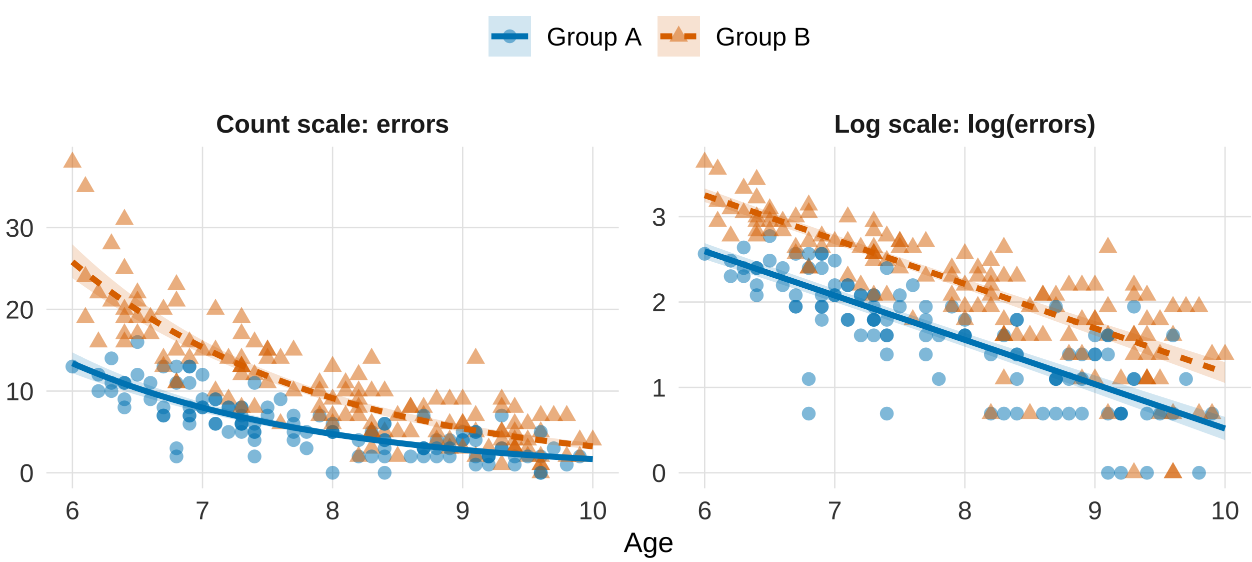
Keywords: interactions, moderation, generalized linear models, link functions, model misspecification

The Link Function Problem in Psychological Interaction Testing

Suppose two groups of children differ in their overall error rates on a task, and errors decrease with age. Figure 1 shows simulated data from such a scenario. In the left panel, errors decline with age in both groups, and the difference between groups is large at age six and nearly closed at age ten. Is there an age-by-group interaction here?

Figure 1

Simulated error counts for two groups of children aged 6 to 10, shown on the observed count scale (left) and on the log scale (right). The two groups have exactly the same age effect on the log of the expected error count, so on the log scale the two fitted lines are parallel, with a constant group gap at every age. Mapped back onto the count scale, the same model produces converging curves, so the group difference shrinks with age. Whether this pattern is an interaction depends on the scale on which effects are assumed to add: on the count scale the group difference changes with age; on the log scale it does not.



Because an interaction is a difference between differences, the question is whether the group difference itself changes with age. On the observed count scale, the group effect does appear to shrink as age increases, so it looks like an interaction. But looking at the log-scale (Figure 1, right panel), both groups have the same age slope, meaning their fitted lines are parallel. A constant difference on the log scale corresponds to a constant ratio on the count scale, so the absolute gap between groups is proportional to the baseline and shrinks as errors decline with age. So whether we call this an interaction is not a fact about the data. It depends on which scale we assume effects add up linearly on. Additive on the log scale means multiplicative (and non-parallel) on the count scale, and vice versa.

This ambiguity extends far beyond this single illustration, as interaction claims are central to psychological theory [**ADD REF**]. Whenever researchers ask whether an effect is moderated, or whether it holds only under certain boundary conditions, they are asserting that the impact of one variable depends on another. This count example therefore demonstrates that the reality of such a claim can shift entirely based on the scale on which those differences are computed. In psychology, this scale dependence has a characteristic source in measurement. Researchers routinely conceptualize constructs such as cognitive ability or symptom severity as continuous latent dimensions [**ADD REF**]. Those dimensions are rarely observed directly; what reaches the data are bounded, discrete quantities, such as counts, ratings, and accuracy scores that carry floors, ceilings, and chance levels. Each measurement scale imposes its own mapping from the latent dimension to the observed response, and that mapping fixes the scale on which effects add.

Generalized linear models (GLMs) make this mapping explicit and give it a standard vocabulary (McCullagh & Nelder, 1989; Nelder & Wedderburn, 1972). A GLM separates two choices that are easy to conflate: the outcome family, which describes the response, and the link function, which sets the scale on which effects combine. Box 1 defines both.

Box 1: Generalized linear models: outcome family and link function

The *outcome family* is the distribution assumed for the response. It defines which values the outcome can take and how observations vary around their expectation. The *link function*, $g(\cdot)$, maps the expected outcome, $\mu_i = E(Y_i)$, onto the linear predictor, η_i, the scale on which the model is additive. With two predictors and their product term,

$$g(\mu_i) = \eta_i, \qquad \eta_i = \beta_0 + \beta_1 X_i + \beta_2 Z_i + \beta_3 X_i Z_i.$$

The inverse link, $g^{-1}(\cdot)$, maps this additive structure back onto the expected response. Its familiar role is to keep expected values within the admissible range of a constrained outcome: $\exp(\eta)$ keeps expected counts positive, and the inverse logit keeps probabilities between 0 and 1. Less obviously, the link thereby defines the scale on which effects add, and therefore what the absence of an interaction, $\beta_3 = 0$, means in the fitted model. The product-term coefficient β_3 is therefore an interaction on the link scale. A response-scale interaction is a different quantity: a difference between expected differences on the observed outcome, for example, whether the group difference in expected errors changes with age in Figure 1. Unless the link is the identity, the two quantities can disagree (Ai & Norton, 2003; McCabe et al., 2022; Mize, 2019). Each family has a canonical link, but other links are legitimate. A Gaussian linear model is the special case that combines a Gaussian family with an identity link, so the additive scale and the response scale coincide and the scale question is invisible. The model behind Figure 1 is a Poisson family with a log link. The family respects that errors are non-negative counts, and the link places additivity on the log scale.

Often the outcome family is chosen with care while the link function is left at its software default, fixing the scale of additivity without deliberation. Two models can share a family and still place additivity on different scales. For example, a Gamma family suits reaction times data; within that family, a log link makes effects additive in log milliseconds, a proportional change, and the canonical inverse link makes them additive in reciprocal means, a change in speed. A binomial family fits correct-versus-incorrect responses and leaves the link open among logit, probit, and chance-corrected forms.

While the family restricts the set of admissible links, model estimates depend heavily on both choices (Hainmueller et al., 2019; Rimpler et al., 2026). When the outcome distribution or the functional form is misspecified, additive structure can appear as a spurious interaction (Ai & Norton, 2003; Busemeyer & Jones, 1983; Czado & Santner, 1992; Domingue et al., 2024; Liddell & Kruschke, 2018; Lubinski & Humphreys, 1990; McCabe et al., 2022; Mize, 2019; Yu & Huang, 2019), and measurement error and low power compound the problem (Baranger et al., 2023; Castillo et al., 2026). That said, an interaction on a scale other than the one implied by the generative model is not a false positive in a strict statistical sense; it is a genuine mathematical property of that scale (Cox, 1984; Domingue et al., 2024; Loftus, 1978; McCabe et al., 2022). Some interactions are invariant to scale; crossover interactions, for instance, survive any monotonic rescaling of the response variable (Cox, 1984; Loftus, 1978; Wagenmakers et al., 2012).

This article addresses these issues in two parts. The first is a preregistered descriptive review of recent empirical articles in five prominent psychology journals, quantifying how often interactions are tested, how often the outcome family and link function are made explicit, and which response formats predominate. The second develops the link function problem in three common response formats: forced-choice accuracy with a chance floor, sum scores, and within-family link choices such as logit versus probit. Using simulations, it quantifies how often an inadequate link turns a purely additive structure into a statistically significant interaction, and whether standard diagnostics detect the misspecification. Table 4 then summarizes these insights into a practical workflow for applied research.

Current Practice in Psychological Interaction Testing: A Preregistered Review

The review examines whether the scale-dependence problem identified above is visible in current reporting practice. We operationalized this through four questions: how often do empirical articles test interactions, how often do interaction-testing articles use non-identity links, how often is the link function explicitly reported, and which response-variable types appear in interaction analyses. The review uses a descriptive coding scheme based on information that readers can usually observe in published reports. Article-level classification of interactions as removable or nonremovable requires more information and a more specific target claim. This classification is clearest in the 2×2 case, where it can often be read from the presence or absence of a crossover. In multi-level, higher-order, or continuous designs, it can depend on the admissible transformations, on the functional form, and on which conditional contrasts or regions of the predictor space carry the substantive claim. Published reports rarely provide all the information needed for this judgment, such as complete coefficients, interaction plots, and the assumed measurement scale. The review therefore documents whether the family, link function, and scale of additivity are made explicit enough for readers to understand which interaction claim was tested (Cox, 1984; Loftus, 1978; Mustillo et al., 2018; Wagenmakers et al., 2012).

Review Method

The review protocol was preregistered before the records were screened, at: https://osf.io/bdu73/ We specified a descriptive review of recently published psychological research articles, with four primary goals: estimating how often empirical articles test interactions, how often interaction-testing articles use non-identity links, how often the link function is explicitly reported, and which response-variable types are used in interaction analyses. The protocol defined a descriptive review of current practice across major areas of psychology, not a census of all journals or publication years.

Sampled articles were recent publications, all from 2024, in five pre-specified reference journals across major areas of psychology: *Psychological Science*, *Journal of Experimental Psychology: General*, *Journal of Personality and Social Psychology*, *Developmental Science*, and *Psychological Medicine*. These journals were chosen to cover general, experimental, social, developmental, and clinical psychology. The sampling design was targeted and descriptive. It was not intended as a random or stratified sample of all psychology journals. Its purpose was to characterize reporting practice in outlets whose standards and reach make them broadly influential beyond the sampled year. Eligible articles were empirical articles presenting original psychological research. Reviews, meta-analyses, editorials, theoretical articles without empirical data, qualitative-only articles, replication studies, and psychometric validation studies were excluded. The preregistered coding scheme targeted observed-outcome analyses and excluded articles in which the relevant tests were conducted only at the latent-variable level, or in which the empirical analyses relied only on structural equation models (SEM). This scope decision kept the review focused on observed outcomes analyzed with regression, ANOVA, or GLM-like models. The screening-flow table records 12 SEM or latent-variable-only exclusions, so this decision affected only a small part of the screened set. SEM is relevant to the broader issue because it can make the measurement model explicit. It also places the link-function problem on a different level, involving thresholds, loadings, and latent-response metrics. Evaluating such models with the observed-outcome coding scheme would have required additional preregistered coding rules. SEM therefore falls outside the scope of this review and would benefit from a dedicated preregistered protocol. Articles were then coded in detail if they tested at least one interaction effect, either through statistical modeling or analysis of variance.

For each interaction-testing article, coders recorded whether at least one tested interaction used a non-identity link function, whether the link function was explicitly reported, whether at least one interaction involved a formally incorrect identity-link analysis of a constrained observed outcome, whether at least one such interaction was statistically significant or interpreted analogously, and which response-variable types were analyzed. We use the term formally incorrect identity-link analysis of a constrained observed outcome for cases in which an interaction was analyzed with an identity link although the reported outcome had formal constraints, such as bounds, discreteness, positivity, or a known chance level. In these cases, the identity link is formally mismatched as a generative model for the observed outcome, unless it is explicitly justified, for example because the target estimand is deliberately defined on the observed-score scale. We use this label as a screening flag for a mismatch between the observed outcome constraints and the identity-link generative model. It does not imply that an observed-score estimand is invalid, or that the substantive conclusion is false. The category groups heterogeneous cases, from mild observed-scale approximations far from the bounds to analyses where the outcome constraints make identity-link additivity much harder to defend. The coding therefore documents the potential scale of the reporting problem, while the worked examples and simulations show mechanisms by which such choices can matter. The protocol specified descriptive summaries, including proportions and Wilson 95% confidence intervals for Boolean variables. Double-coded records with complete coder pairs showed high agreement, and the same information is reported together with Cohen’s kappa: 92.0% agreement, kappa = 0.81

for non-identity link use, 88.9% agreement, kappa = 0.66 for explicit link reporting, 88.3% agreement, kappa = 0.71 for formally incorrect identity-link analyses of constrained observed outcomes, and 91.5% agreement, kappa = 0.68 for significant interactions among these cases. We report raw agreement and kappa together because several coded variables had imbalanced marginal distributions. The coder-1/coder-2 yes rates were 28.1% / 30.2% for non-identity link use, 20.6% / 20.6% for explicit link reporting, 71.6% / 72.1% for formally incorrect identity-link analyses, and 86.5% / 82.3% for significant interactions among these cases. These marginal rates document why kappa can be conservative in this setting, while still allowing readers to evaluate the agreement pattern directly. The full review protocol, coding definitions, and extended descriptive tables are reported in the Supplement.

Review Findings

Three descriptive results organize the review findings. First, interaction testing was common. The review identified 331 eligible empirical articles, and 200 of these articles tested at least one interaction, corresponding to 60.4%, 95% CI [55.1, 65.5]. This confirms that interaction testing is a routine part of empirical reporting in the sampled journals.

Second, model-scale reporting was uncommon. Among the 200 interaction-testing articles, 58 used at least one non-identity link function, 29%, 95% CI [23.2, 35.6], and only 46 explicitly reported the link function, 23%, 95% CI [17.7, 29.3]. Thus, in about 77.0% of interaction-testing articles, the link function was not explicitly stated. This is the main reporting problem documented by the review: the scale on which no interaction is assumed is often left implicit.

Third, constrained outcomes were often analyzed with identity-link models. Among the 200 articles testing interactions, 144 included at least one identity-link analysis in which the reported outcome had formal constraints that make identity-link additivity formally incorrect as a generative model unless explicitly justified as an approximation or as the target observed-score metric, 72%, 95% CI [65.4, 77.8]. This proportion is a conservative screening count. It is not an estimate of false substantive conclusions, and it is not a severity rating. The category uses transparent screening and avoids retrospective case-by-case severity judgments that published reports often could not support. Cases were flagged only when the formal constraint was identifiable from the published report. When the available report did not allow us to establish this mismatch, the case was not counted. The review result shows that many published reports did not make the family, link, and scale-of-additivity choices explicit enough for readers to evaluate the interaction claim. This is the same broad concern raised by work on outcome-model mismatch and metric assumptions in interaction testing (Domingue et al., 2024; Liddell & Kruschke, 2018). Among these 144 cases, 124 reported at least one statistically significant interaction, 86.1%, 95% CI [79.5, 90.8]. In absolute terms, this corresponds to 124 of the 200 interaction-testing articles, or 62.0%. This last number should not be read as a false-positive estimate. It records how often formally incorrect identity-link analyses appeared in reports where statistically significant interaction claims could matter for interpretation. Table 1 summarizes these descriptive results.

Table 1

Descriptive summary of the preregistered review. Formally incorrect identity-link analyses of constrained observed outcomes are cases where the reported outcome had formal constraints that make identity-link additivity formally incorrect as a generative model unless explicitly justified as an approximation or as the target observed-score metric.

Quantity	n	Denom.	%	95% CI
Eligible empirical articles	331	331	100.0%	[98.9%, 100.0%]
Eligible empirical articles testing at least one interaction	200	331	60.4%	[55.1%, 65.5%]
Interaction-testing articles using at least one non-identity link	58	200	29.0%	[23.2%, 35.6%]
Interaction-testing articles explicitly reporting the link function	46	200	23.0%	[17.7%, 29.3%]
Interaction-testing articles with formally incorrect identity-link analyses of constrained observed outcomes	144	200	72.0%	[65.4%, 77.8%]
Formally incorrect identity-link cases with at least one significant interaction	124	144	86.1%	[79.5%, 90.8%]
Eligible empirical articles not testing interactions	131	331	39.6%	[34.5%, 44.9%]

The pattern was present across the sampled journals. In the supplementary by-journal table, the proportion of formally incorrect identity-link analyses of constrained observed outcomes among interaction-testing articles ranged from 40.0% in Psychological Medicine to 95.0% in Psychological Science. In this sample, Psychological Medicine had the lowest screening proportion and the highest non-identity-link proportion (40.0%). This journal, which uses many outcomes for which non-identity links are standard, contributed the fewest formally incorrect identity-link cases. This pattern also shows that the overall estimate was not driven by the clinically oriented journal, where non-identity links were more common.

The outcomes used in interaction analyses were heterogeneous. The most common broad, non-exclusive categories were binary, accuracy, or proportion outcomes (84 articles), sum scores or composites (66 articles), ordinal, Likert, or rating outcomes (36 articles), response times or durations (30 articles), neural or physiological measures (23 articles), difference or distance scores (17 articles), counts or error counts (9 articles), and correlations or associations (8 articles). This heterogeneity matters because the identity link has different implications across outcome types. It also shows why the worked examples focus on bounded, binomial, ordinal, and aggregated responses. Many observed psychological variables are built from binary or ordered response events, then reported as proportions, ratings, averages, totals, or composites. For some approximately continuous outcomes, additivity on the observed scale may be a defensible target. For bounded, discrete, ordinal, chance-constrained, or heavily transformed outcomes, however, additivity on the observed scale is an assumption that should usually be stated and examined. This is especially clear for ordinal and Likert-type outcomes, where treating ordered categories as metric can affect estimates, error rates, and interaction conclusions (Bürkner & Vuorre, 2019; Liddell & Kruschke, 2018).

What the Review Shows, and What It Does Not

The review shows that while interaction testing is pervasive in leading psychology journals, both the outcome family and link function are frequently left implicit, leaving readers to guess which specific interaction claim was actually tested.

Table 2

Broad response-variable categories in interaction-testing articles. Categories are non-exclusive, so one article can contribute to more than one row.

Outcome type	n	Denominator	Percent
Binary / accuracy / proportions	84	200	42.0%
Sum scores / composites	66	200	33.0%
Ordinal / Likert / ratings	36	200	18.0%
Response times / durations	30	200	15.0%
Counts / error counts	9	200	4.5%
Neural / physiological measures	23	200	11.5%
Correlations / associations	8	200	4.0%
Difference / distance scores	17	200	8.5%

Two limits define what the review cannot show. First, the review documents how analyses were reported. An identity-link analysis of a constrained outcome can be a deliberate and defensible choice when the estimand is defined on the observed scale, so the flagged cases are reporting gaps. Second, a descriptive review cannot establish how often published interactions are scale artifacts, because that judgment requires knowing the scale on which the data-generating process is additive. The remainder of the paper therefore turns to settings where the generating process is known by construction.

The Link Function Problem in Three Common Response Formats

To demonstrate the practical consequences of link misspecification, we develop three complementary simulation studies:

1. *Forced-Choice Accuracy*: This case holds the binomial outcome family fixed and evaluates whether the link function must explicitly encode a task-implied chance floor (e.g., a 0.50 guessing threshold).
2. *Within-Family Link Selection (Logit vs. Probit)*: This case holds both the binary family and the 0-to-1 bounds fixed, testing whether choosing between these two standard link functions alters the results.
3. *Sum Scores*: This shows how adding up individual item responses to create a bounded total

score blends two distinct issues: how individual items map onto a total scale, and how that scale alters what it means for an effect to be truly additive.

Each case develops the conceptual problem and then reports a simulation that quantifies it. Across the three cases, the simulations address two questions: first, whether a purely additive data-generating process on one scale is detected as a statistically significant interaction when modeled on a different scale; and second, whether standard post-estimation diagnostics reliably identify the misspecified link. The second question is taken up in a dedicated diagnostics section after the three cases.

The simulation scenarios were chosen to represent plausible psychological data patterns. Each reproduces a common interaction-testing situation: two effects that are already well established, for example a group difference combined with a developmental trend or a condition effect, and the question of whether one effect moderates the other. The two main effects were always set strong enough to push expected values toward a floor, ceiling, or chance level while keeping them clearly inside the observed range. This is the region where link curvature acts most strongly, and it corresponds to patterns that psychological studies routinely report, such as accuracy that approaches ceiling in one group and stays near chance in another. Every constraint that defines a scenario, such as a .50 guessing floor or a bounded total score, can be stated from the task and the scoring rule before seeing the data.

Each simulation applies the same design. Data are generated with no product-term interaction on the generating scale; models assuming additivity on the generating scale and on plausible alternative scales are fitted to the same data; and the proportion of statistically significant product terms across replications quantifies the inferential risk attached to each scale choice. The reported rates therefore quantify this mechanism under known generating processes, and they are specific to these tuned scenarios.

Forced-Choice Accuracy and the Non-Zero Chance Floor

If a participant answers Y_i of k_i trials correctly, a binomial sampling model, $Y_i \sim \text{Binomial}(k_i, p_i)$, is the natural starting point. The family fixes the sampling process; it does not fix the lower bound that is meaningful for the task. In an m-alternative forced-choice task, guessing yields an expected accuracy of $c = 1/m$, which is .50 in a two-alternative or true/false task, so a claim about performance above guessing places the lower asymptote of the performance curve at c, not at 0.

Three specifications treat this floor differently. A Gaussian identity model on the raw proportions ignores the trial process, assumes additivity in percentage-correct units, and can predict outside the 0-to-1 range. A standard binomial logit or probit model, $p_i = F(\eta_i)$ with F the logistic or standard-normal CDF, respects the sampling process and the 0-to-1 bounds and still places the lower asymptote at 0. A chance-corrected binomial link keeps the binomial family and maps the linear predictor onto the interval from chance to 1,

$$p_i = c + (1 - c)F(\eta_i),$$

which for $c = .50$ becomes $p_i = .50 + .50\,F(\eta_i)$; equivalently, the link acts on the above-chance quantity $(p_i - c)/(1 - c)$. A product term in the standard model tests additivity over the full 0-to-1 range, and a product term in the chance-corrected model tests additivity on a performance-above-chance scale that already encodes the task floor.

The mismatch is most severe when one group or condition lies near chance. Suppose age has the same effect in two groups on the performance-above-chance scale, with no true age-by-group product term there. Near .50 the chance-corrected curve has little response-scale room, so the lower-performing group shows stronger compression on the observed accuracy scale. A standard logit or probit link puts that compression near 0 in place of near .50, and the fitted product term absorbs the residual curvature. The age-by-group difference between differences is exactly zero on the chance-corrected scale and systematically nonzero on the standard logit scale, so the standard model detects a genuine property of its own scale, and detects it more reliably as data accumulate.

Standard binomial models remain useful for many accuracy analyses. The chance-corrected link is most defensible when the chance level is known from the task design, guessing is a plausible lower-bound process, and below-chance performance is not substantively meaningful for the target claim. In the simplest case, the chance level can be fixed by design, for example $c = 0.50$ in a 2-AFC task. In other cases, analysts may estimate the lower asymptote or report sensitivity analyses over plausible values of c; estimating it requires enough information and stronger assumptions. Richer models may also be needed if there are lapses, response biases, item heterogeneity, learning across trials, systematic below-chance performance, or participant-specific guessing strategies.

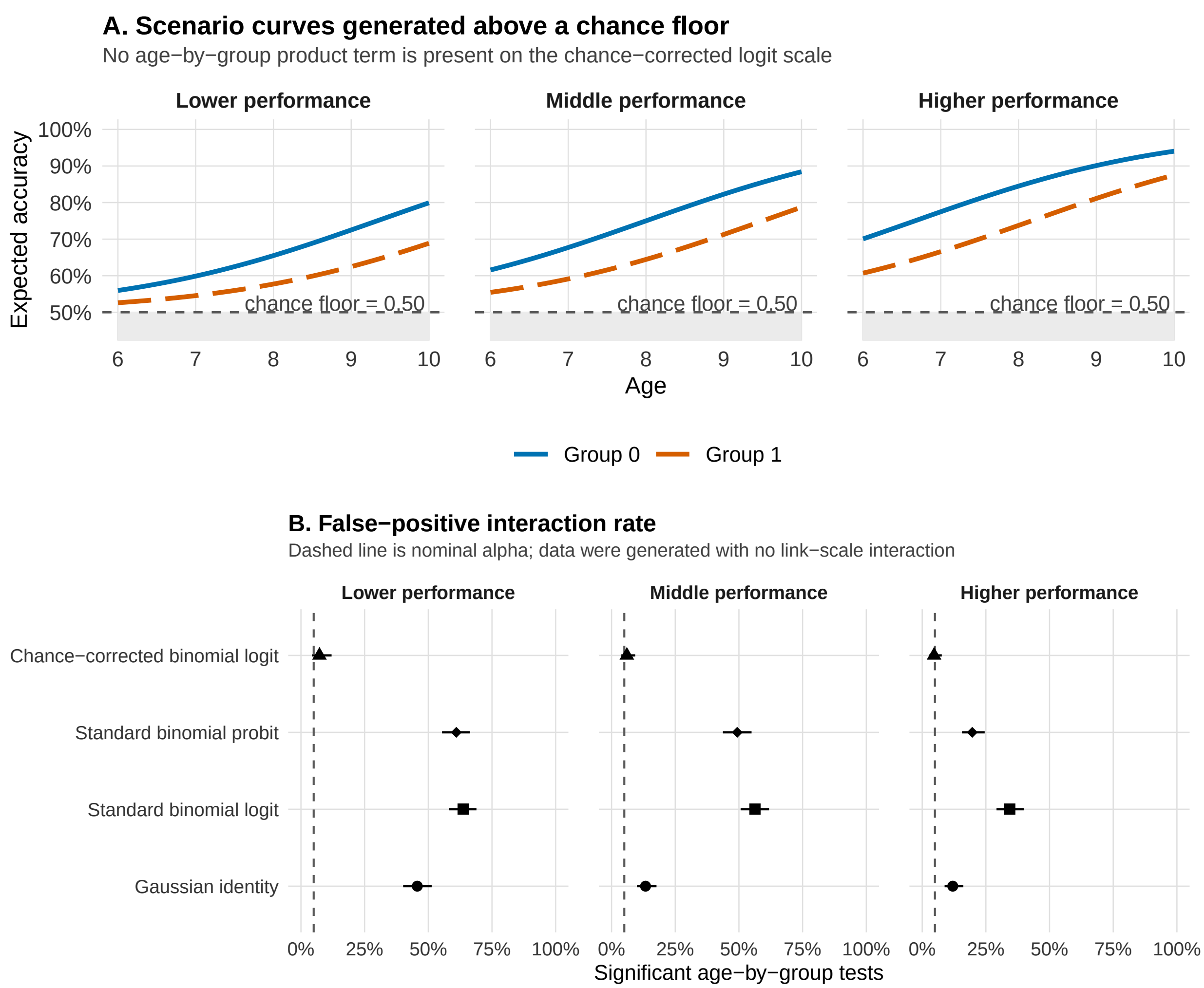
The chance-corrected GLM link is closely related to psychometric-function models (Knoblauch & Maloney, 2012; Wichmann & Hill, 2001). For example, the 3-parameter logistic IRT model includes a guessing parameter at the item level, so the chance process is modeled where it operates. When item-level or trial-level data are available, and when the target claim concerns latent performance rather than an aggregated accuracy proportion, such measurement models or trial-level generalized linear mixed models (GLMMs) may be more appropriate than a single aggregate GLM (Moscatelli et al., 2012). The chance-corrected link used here should therefore be understood as a GLM-level representation of a design constraint. It is not a replacement for item-level measurement models when the data and the scientific question call for them.

Simulation 1: Design and Results

In each of 300 replications we generated accuracy for $N = 250$ participants with $k = 20$ trials each, from a chance-corrected logit model with a .50 floor, a positive age effect, a group difference, and no age-by-group product term on the generating scale. Three scenarios differ only in overall performance, set by shifting the intercept: in the *lower-performance* scenario the two groups lie close to the .50 floor (expected accuracies roughly .53 to .80 across ages 6 to 10), in the *middle-performance* scenario they occupy the middle of the above-chance range (roughly .55 to .88), and in the *higher-performance* scenario they approach the ceiling (roughly .61 to .94). Each simulated dataset was analyzed with four aggregate models, one row per participant: a Gaussian linear model on the observed proportions, standard binomial-logit and binomial-probit GLMs on the correct-response counts, and the chance-corrected binomial-logit GLM that matches the generating scale. We recorded how often each returned a significant age-by-group product term at $\alpha = .05$, a false positive by construction. Figure 2 shows the simulation results, also summarized in Table A1 in the supplement.

Figure 2

Simulation 1: forced-choice accuracy with a task-implied chance floor. Panel A shows the deterministic scenario curves generated by the chance-corrected logit model. The grey region marks values below the .50 chance floor, and the dashed horizontal line marks chance performance. Panel B shows the false-positive rate for the age-by-group product term across replications, for each fitted model and performance scenario. Data are generated with no age-by-group interaction on the chance-corrected scale.



The chance-corrected model stays near the nominal .05 in every scenario, with false-positive rates between 5% and 7%, because it matches the generating scale. The misspecified models inflate, most severely where performance lies near the floor: in the lower-performance scenario the standard logit rejects 64% of the time, the standard probit 61%, and the Gaussian identity 46%, against the nominal 5%. The inflation shrinks as performance moves away from the floor, falling in the higher-performance scenario to 34% for the logit and 20% for the probit, and it persists wherever the inverse link is curved. When the true floor is .50 and the fitted link places it at 0, the product term compensates for the wrong no-interaction baseline.

The chance-corrected model can be numerically demanding. Near the floor the likelihood carries little information about changes on the above-chance scale, and some fits fail to converge; in the lower-performance scenario, only 179 of the 300 replications did. This instability is itself informative, signaling that the design gives limited evidence about above-guessing differences in that region. Weakly informative Bayesian priors or item-level models can stabilize estimation, at the cost of added assumptions that should be stated and checked [**REF O TOGLIERE FRASE**].

Within-Family Link Choice

The forced-choice case involved a link that was plainly wrong for the task, placing the lower asymptote at 0 when the design fixed it at chance. The link-function problem also takes a subtler form, between two links that are both standard for binary data, both respect the 0-to-1 bounds, and usually produce almost indistinguishable fits. Logit and probit give nearly identical fitted probabilities across most of the response range, which is why switching between them leaves most applied conclusions unchanged. That practical equivalence makes the pair a stringent test of the link-function problem, because a distortion of an interaction even here would extend well beyond obvious misspecification. But the equivalence is only local: near the floor, ceiling, chance level, or distributional tails, the two links imply different distances on the link scale, even where their fitted probabilities nearly.

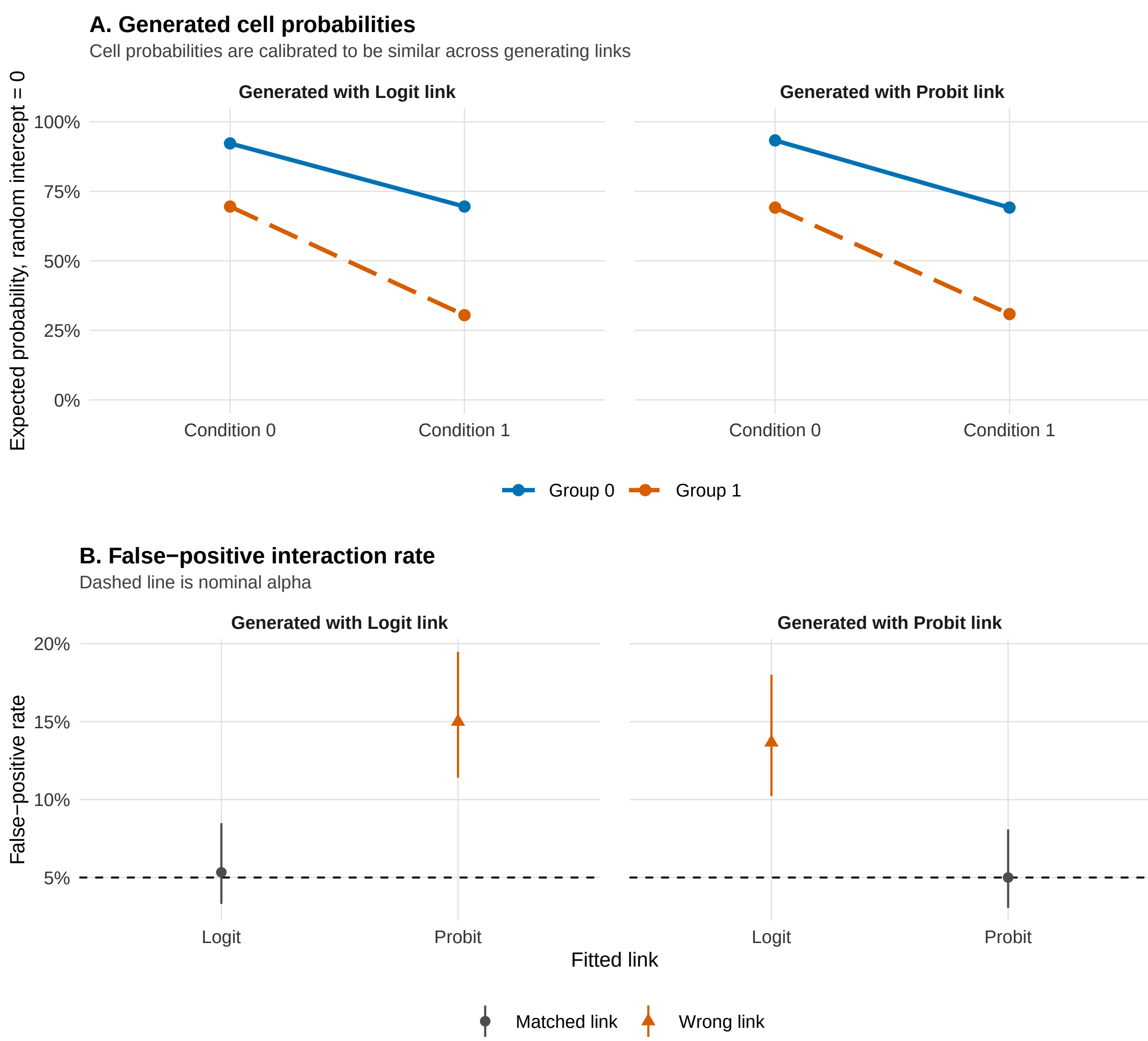
Additionally, the two links still encode different substantive interpretations. A logit link is natural when the interpretation is in odds or odds ratios: additivity on the linear predictor scale then means additivity in log odds, and exponentiated coefficients have a direct odds-ratio interpretation. A probit link is natural when the response can be represented as threshold crossing on an approximately normal latent variable, such as ability, signal strength, response tendency, propensity, or latent correctness (Dunn & Smyth, 2018).

Simulation 2: Design and Results

In each of 300 replications we generated repeated binary responses for $N = 700$ participants in a two-group, two-condition design, with 15 trials per participant-condition cell and a subject random intercept (latent intraclass correlation 0.30). The condition-by-group product term was zero on the generating link scale. We crossed the generating link with the fitted link, using logit and probit in both roles, and fitted every model as a random-intercept GLMM, recording how often each returned a significant condition-by-group product term at $\alpha = .05$, a false positive by construction. Figure 3 shows the results, also summarized in Table A2 in the supplement.

Figure 3

Simulation 2: within-family link choice in repeated-trial random-intercept GLMMs. Panel A shows the deterministic cell probabilities generated by additive logit or probit models at random intercept 0. Panel B shows the false-positive rate for the condition-by-group product term when the generating and fitted links are crossed. Error bars are Wilson 95% confidence intervals, and the dashed line marks nominal alpha.



When the fitted link matches the generating link, the false-positive rate stays near the nominal .05: 5% for logit fitted to logit-generated data and 5% for probit fitted to probit-generated data. Crossing the links inflates it to 14% when a logit model is fitted to probit-generated data and 15% when a probit model is fitted to logit-generated data, roughly a tripling of the nominal rate. The inflation is smaller than in the forced-choice case, and it appears among links that share the same family, the same 0-to-1 bounds, and nearly overlapping fitted curves in the middle of the probability range, the setting many applied analysts treat as interchangeable. Under the wrong link the same cell probabilities imply a small but deterministic nonzero product term on the fitted scale, which repeated sampling detects. A result that is stable across both links is easier to defend; one that appears only under a single link should be reported as link-conditional, or translated into a response-scale estimand and checked across plausible links.

Sum Scores as Bounded and Discrete Outcomes

Sum scores often combine a link-scale issue with a target-metric issue. They may be analyzed as Gaussian identity outcomes, transformed bounded scores, ordinal summaries, or item-level response models, depending on the scientific target and on the information available. We include them because many sum scores aggregate binary or ordinal response processes. For interaction claims, the risk is the same: the chosen metric or link defines the scale on which additivity is tested.

Sum scores make the link-function problem inseparable from measurement. Many psychological outcomes are reported as totals or averages across questionnaire items, task items, symptoms, errors, or ratings. Recent debates on sum scores make the same point from a psychometric angle: a total score can be useful and transparent, but it also encodes assumptions about how item responses represent the target construct (McNeish, 2023; McNeish & Wolf, 2020; Widaman & Revelle, 2023). These scores are often treated as if they were direct observations of an approximately continuous psychological dimension. The latent construct and the observed score are different objects. The construct may be conceived as a continuous dimension, and may sometimes be reasonably approximated as smooth or roughly normal. The sum score is produced by a finite set of item responses through thresholds, endorsement probabilities, success probabilities, category labels, and scoring rules. It is therefore a remapped version of the target continuum.

This point matters even when the construct is otherwise well behaved. A normally distributed latent variable can still generate a bounded, discrete, skewed, or non-interval observed score. A total based on binary or ordinal items has only a limited number of possible values. It has a lower and upper bound. It can be skewed when items are too easy, too hard, rarely endorsed, or almost always endorsed. It can be compressed near floor or ceiling, but compression is not restricted to the endpoints. Adjacent raw-score values may correspond to different distances on the latent scale at different regions of the score range. These features are common in psychological variables (Micceri, 1989), and they are central to why treating ordinal responses as metric can produce false alarms, missed effects, reversals, and misleading interactions (Liddell & Kruschke, 2018).

Sum scores can be useful, transparent, reliable enough for many purposes, comparable across studies, and pragmatically tied to how instruments are scored in applied settings (Widaman & Revelle, 2023). A Gaussian identity model for a sum score still makes a scale claim. It assumes that effects are additive in raw-score units on the observed score scale. In an interaction model, the no-interaction baseline is that the expected raw-score effect of one predictor is constant across levels of the other predictor. Sometimes raw-score units are the target estimand. For example, a clinician, teacher, or researcher may care about a two-point difference on the reported score because decisions and interpretations are made on that metric.

Gaussian identity models are more defensible when the score has many possible values, the distribution is not strongly skewed, observations are not concentrated near floor or ceiling, residual patterns are not strongly tied to fitted values, and the substantive claim is explicitly about raw-score differences. They are less defensible when the score has few levels, is built from ordinal item responses with unclear spacing, shows floor or ceiling concentration, or is interpreted as evidence about moderation on a latent construct scale.

The problem becomes sharper when the verbal claim moves from the manifest score to the underlying construct. A group near the upper bound may show a smaller observed treatment effect simply because the score has less room to increase, with the treatment equally effective on the construct. A group located near a dense region of thresholds may show a larger raw-score change because the instrument is more sensitive there. The product term can then partly reflect the target metric rather than genuine moderation in the psychological process. A closely related problem has been shown in item-level work on genotype-by-environment interactions, where summed item scores can mask genuine interactions or generate spurious ones through scale properties of the observed score (Molenaar & Dolan, 2014). This is the sum-score version of outcome-model misspecification and target-metric mismatch (Domingue et al., 2024).

Several pragmatic alternatives can make the scale assumption more explicit. For a manifest sum score, a Gaussian identity model can be a defensible approximation when the claim is explicitly about raw-score units, the score has many levels, predicted values remain in range, and floor or ceiling compression is limited. When the claim is closer to an underlying construct, the better solution is an explicit measurement model, such as item-level ordinal, IRT-inspired, or SEM models. A bounded-score probit model can be used only as a sensitivity analysis on the manifest total. It is not a substitute for a measurement model. This point is also consistent with the sum-score debate: replacing a total score with a latent-score or item-level model moves the assumptions into the measurement model (McNeish, 2023; McNeish & Wolf, 2020; Widaman & Revelle, 2023). The role of the alternatives considered here is not to establish the true scale, but to test whether the interaction conclusion depends on the manifest sum-score metric (Bürkner & Vuorre, 2019; Domingue et al., 2024; Liddell & Kruschke, 2018).

The recommendation is to make the assumed scale visible. Researchers can inspect the score distribution, plot predicted values near the bounds, report response-scale contrasts, and compare Gaussian identity results with bounded-score probit, ordinal, bounded-count, or item-level alternatives when these are plausible. Stable interactions across these specifications are easier to defend. Interactions that appear only on the manifest sum-score metric can still be informative, but they should be framed as properties of that score and interpreted as construct-level moderation only with supporting measurement assumptions.

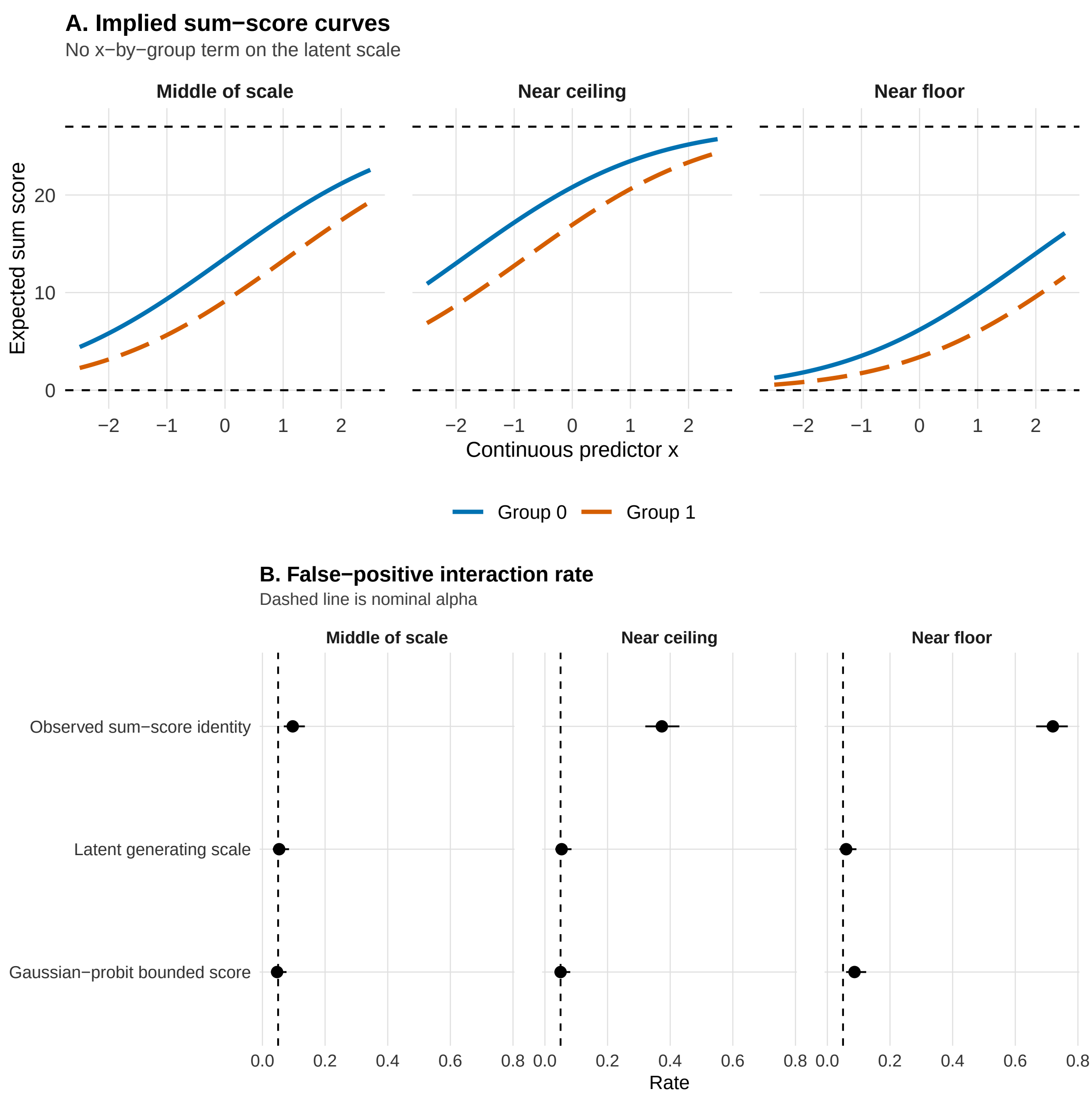
Simulation 3: Design and Results

In each of 300 replications we generated data for $N = 600$ participants on a latent continuous scale, with a continuous predictor, a group effect, and no true x-by-group product term. Each latent value was converted into 9 ordinal items scored 0 to 3, which were summed to a total on a 0-to-27 scale. Three scenarios shift the item thresholds so that expected totals fall in the middle of the scale, near the floor, or near the ceiling. Each participant's total score was analyzed on three scales, chosen to isolate the relevant comparison: the observed sum-score identity model, a Gaussian linear model of the manifest total; the Gaussian-probit bounded-score model, a sensitivity model that maps the total to a bounded proportion through a probit link; and the latent generating scale itself, an idealized benchmark available only because the true theta is known. We recorded how often each returned a significant x-by-group product term at $\alpha = .05$, a false positive by construction. Figure 4 shows the results, also summarized in Table A3 in the supplement.

The bounded-score model fits a continuity-corrected score proportion with a Gaussian likelihood and a probit link, then rescales predictions to the sum-score metric. The probit link is more interpretable here than a logit link because its linear predictor is on a standard-normal, z-like scale. In the idealized threshold-normal setting used in the simulation, this makes the coefficient scale closer to standardized latent differences. This interpretation remains conditional on the measurement assumptions and should not be read as a literal Cohen's d for the target construct.

Figure 4

Simulation 3: sum scores as bounded and discrete outcomes. Panel A shows how a latent additive data-generating model, with no x-by-group product term, is mapped onto the observed sum-score scale under middle, floor, and ceiling scenarios. Panel B shows false-positive interaction rates across repeated simulations. The latent generating scale is an idealized benchmark available only because theta is known in the simulation; applied analyses have no access to it.



The latent generating scale, the idealized benchmark, stays near the nominal .05 in every scenario (between 5% and 6%), confirming that the interaction is genuinely absent on the scale that generated the data. The observed sum-score identity model matches it only in the middle of the scale, where the total is approximately linear in the latent variable (10%). As expected totals approach a bound, the raw-score model inflates sharply, reaching 37% near the ceiling and 72% near the floor, because uneven compression of the latent scale makes a constant latent group effect appear to change across levels of the predictor. The Gaussian-probit bounded-score model stays close to nominal throughout (between 5% and 9%), showing that respecting the bounds of the score removes most of the inflation.

The pattern is conditional on the region of the score range. Far from the bounds, and for a claim about raw-score units, the Gaussian identity analysis of the total approximates the latent-scale result. Near a bound, the raw-score no-interaction baseline diverges from the latent one, and a product term can appear that reflects compression of the metric even when the construct-level effect is constant. The bounded-score probit model is a useful sensitivity check, and it remains a model of the manifest total. When the claim concerns an interaction between latent constructs, the more direct solution is an explicit measurement model, such as SEM, item-level ordinal, or IRT-inspired models; these still require assumptions about thresholds, links, loadings, and latent-response metrics, so they make the measurement part of the interaction claim explicit without removing the scale problem.

What Diagnostics Can and Cannot Detect

Diagnostics belong in link-aware modeling, and by themselves they leave the link-function question undecided. The central problem is diagnostic misalignment: a wrong link can generate a pseudo-interaction that becomes statistically detectable, while the checks used to diagnose the model remain quiet, have limited power, or flag only broad misfit. In that situation, the interaction test can become sensitive to the wrong scale before the diagnostic evidence becomes specific enough to identify its source.

Pregibon-style link tests target link adequacy directly (Pregibon, 1980). The idea is simple. The analyst fits the model, saves the fitted linear predictor $\hat{\eta}$, and refits the model with $\hat{\eta}^2$

added as a predictor in this secondary analysis. Under an adequate link, the squared term captures nothing systematic and its test stays near the nominal level; a significant squared term signals curvature that the assumed link leaves unexplained. A significant added term therefore suggests that the assumed link may be inadequate, and it stops short of identifying the correct link: the same signal can reflect a missing nonlinear predictor, an omitted covariate, an omitted interaction, or another problem in the systematic component. A non-significant result is also limited: it means that this diagnostic, in this sample, did not detect this particular departure.

Simulation-based residual diagnostics provide a broader check. The DHARMa package implements randomized quantile residuals (Dunn & Smyth, 1996; Hartig, 2024). The idea is again simple. Many new datasets are simulated from the fitted model, and each observed response is located within the distribution of its own simulated values: an observation at the median of its simulated distribution receives a residual of .50, and an observation in the far tail receives a residual near 0 or 1. When the model is adequate, these positions are uniformly distributed between 0 and 1 for every outcome family, so misfit appears as departures from uniformity, as over- or underdispersion, or as residual trends over fitted values and predictors. These residuals are easier to interpret across GLMs and GLMMs than raw, Pearson, or deviance residuals. In the diagnostic simulation below, we use four DHARMa checks: residual uniformity, dispersion, residual quantile patterns over fitted values, and residual structure over the focal predictor, implemented as a quantile check for continuous predictors and as a design-cell comparison for the categorical 2×2 case. These checks detect broad misfit, dispersion problems, and structured residual patterns. By themselves, they do not distinguish link misspecification from family misspecification, a missing nonlinearity, a missing interaction, or a target-metric mismatch.

Model comparison tools, such as AIC, BIC, WAIC, or LOO, add evidence within a clearly defined candidate set. For link diagnostics, their cleanest use is to compare candidate links while holding fixed the observed outcome, likelihood family or support, and structural formula. In that use, AIC can compare Poisson-log and Poisson-identity fits to the same counts, binomial-logit and binomial-probit fits to the same successes and trials, or Gamma-log and Gamma-inverse fits to the same positive outcome. It can also compare a standard binomial link with a chance-corrected binomial link when both define likelihoods for the same binomial data. It is less direct when the comparison changes the outcome, the likelihood support, or the measurement metric, which is why we do not treat sum-score alternatives as a simple AIC contest among links. More importantly, information criteria compare predictive fit among the models supplied by the analyst. They do not decide which scale defines the scientific no-interaction claim. A wrong or incomplete link may even be favored when the product term absorbs part of the curvature created by the misspecification. Thus, a lower AIC is evidence about relative fit within the candidate set; the adequacy of the link for the scientific claim requires separate justification.

For interaction claims, diagnostics therefore contribute one part of the argument. A useful order is: first identify constraints from the task, scoring rule, and measurement process; then define a small set of plausible outcome families, links, and target metrics; then use residual diagnostics, link checks, calibration, and information criteria as supporting evidence; finally, report whether the interaction conclusion is stable across the plausible specifications. A quiet diagnostic result should not end the assessment when the outcome has known constraints. For example, in a forced-choice task with a known chance level, a standard binomial logit can fit many aspects of the data while still encoding the wrong lower asymptote for the scientific claim. If the claim concerns response-scale moderation, researchers should report response-scale contrasts or marginal effects and examine whether those quantities are stable across plausible links (McCabe et al., 2022; Mize, 2019; Rohrer & Arel-Bundock, 2026). If the claim concerns additivity on a latent or linear-predictor scale, the chosen link needs a substantive justification.

Diagnostic Simulation

Finally, we examined whether common diagnostics flag the same link problems that produced the pseudo-interactions. The simulation asks a focused question: does diagnostic evidence appear before, after, or at the same time as the false interaction?

We used five scenarios, each tied to a previous example or matched counterpart in the paper. The first is tied to the simple count example in Figure 1. Data are generated from a

Poisson-log model in which expected errors decline with age, with no age-by-group product term on the log-count scale. The diagnostic scenario uses the same qualitative structure as the figure, with coefficients recorded in the generated scenario table: intercept 2.20, age effect -0.55 per year after age 6, and group effect 0.45. The fitted model keeps the Poisson family and the interaction formula and uses an identity link. This makes the comparison a same-family link comparison: the family and the formula are held fixed, and only the link changes.

The second scenario is the lower-performance forced-choice scenario from Simulation 1. Data are generated with a .50 chance floor, 20 trials per participant, intercept -0.80, age effect 0.60, group effect -0.90, and no age-by-group product term on the chance-corrected logit scale. The fitted model is a standard binomial-logit interaction model, which respects the binomial trial structure while placing the lower asymptote at 0, below the task-implied chance level.

The third scenario mirrors the probit-generated, logit-fitted case in Simulation 2. Data are repeated binary trials with a subject random intercept, 15 trials per condition cell, and latent ICC .30. The fixed effects are the probit reference values used in Simulation 2: intercept 1.50, group effect -1.00, condition effect -1.00, and no group-by-condition product term. The fitted model is the corresponding random-intercept logit GLMM with the same interaction formula.

The fourth scenario is a continuous-predictor counterpart to the previous binary repeated-trials case. It keeps the probit data-generating link, the logit fitted link, the group effect, the focal-predictor effect, the subject random intercept, and the latent ICC, and replaces the two-level condition with a continuous predictor sampled from 0 to 1. This scenario asks whether the same logit-versus-probit issue is visible when the focal predictor has enough unique values for continuous-predictor residual checks to be better identified.

The fifth scenario uses the Gamma response-time example from the introduction. Data are generated from a Gamma model with a log link, shape 8, baseline mean 400 when x = 0 and group = 0, x effect 0.25 on the log-mean scale, group effect 0.25, and no x-by-group product term. The fitted model keeps the Gamma family and the interaction formula and uses the inverse link.

In each replication, diagnostics were applied to the misspecified interaction model, because this is the model an analyst would usually inspect after finding the interaction. This design targets a realistic post-estimation question: does the same fitted model that produced the product-term finding also show enough diagnostic evidence to warn the analyst about the link problem? The choice is demanding because the product term can absorb part of the curvature created by the wrong link. A diagnostic workflow applied to an additive model before adding the product term would answer a different question. The present simulation asks whether ordinary checks protect the interaction conclusion once the misspecified interaction model has been fitted.

The residual workflow used four DHARMa checks: residual uniformity, dispersion, residual quantile patterns over fitted values, and residual structure over the focal predictor, using a quantile check for continuous predictors and a design-cell comparison for the categorical 2×2 case (Dunn & Smyth, 1996; Hartig, 2024). We also used a Pregibon-style added-term link check as a more link-oriented diagnostic (Pregibon, 1980).

Finally, we used AIC only for relative comparisons among prespecified candidate mean models fitted to the same response data with the same interaction formula. In the count and Gamma scenarios, the response family is held fixed and only the link changes. In the binary scenarios, the binomial sampling model is held fixed. The chance-floor scenario compares two binomial mean mappings: a standard logit model and a chance-corrected logit model with a design-implied lower asymptote. Thus, AIC varies only the candidate mean-link mapping, and the product term is present in every candidate model. We report the proportion of replications in which the lower AIC favored the target model. Because no inconclusive ΔAIC category is used, the complement favors the misspecified model when both AIC values are available. These values describe relative fit within a limited candidate set.

Table 3 documents diagnostic misalignment: the same misspecification that generates a pseudo-interaction often does not produce a clear diagnostic signal. In several same-family scenarios, AIC or the Pregibon-style check favored the target model; protection was uneven and depended on which candidate models were compared. The starkest result is the chance-floor scenario: a psychologically common link problem produced a large pseudo-interaction rate while

Table 3

Diagnostic simulation across five link-misspecification scenarios. FP interaction reports how often the misspecified interaction model detected a product term absent on the generating link scale. DHARMa gives the range across the four prespecified residual checks. Pregibon reports the added-term link check. AIC favors target reports the proportion of replications in which the lower AIC favored the target interaction model over the misspecified interaction model, with the structural formula held fixed. The complementary proportion favors the misspecified model when both AIC values are available. CI = Wilson 95% confidence interval.

Case	Target model	Misspecified model	FP interaction	DHARMa	Pregibon	AIC favors target
Count errors	Poisson log	Poisson identity	0.942 [0.908, 0.964]	0.007-0.906 (4)	0.996	0.996
Chance-floor accuracy	chance-corrected binomial logit	standard binomial logit	0.617 [0.561, 0.670]	0.000-0.070 (4)	0.303	0.000
Binary repeated trials	binomial probit GLMM	binomial logit GLMM	0.080 [0.054, 0.116]	0.000-0.020 (4)	1.000	0.923
Binary continuous predictor	binomial probit GLMM	binomial logit GLMM	0.060 [0.038, 0.093]	0.000-0.007 (4)	1.000	0.960
Gamma mean response time	Gamma log	Gamma inverse	0.287 [0.238, 0.340]	0.000-0.043 (4)	0.303	0.777

all four DHARMa checks remained near nominal levels and AIC favored the misspecified model.

Practical Starting Points for Link-Aware Interaction Testing

The applied output of the framework is a small set of outcome-specific starting points. These starting points are prompts for the choices that should be visible in a scientific paper: what family respects the observed response, what link defines the scale of additivity, and what sensitivity check is feasible for the design (Table 4). The preferred link should be justified from theory, task structure, or measurement whenever possible (Baranger et al., 2023; Castillo et al., 2026). When these considerations do not uniquely determine the link, the interaction should be probed through a planned link-sensitivity analysis. Researchers should fit a set of theoretically defensible alternative specifications, compute the same target response-scale contrast across them, and state whether the interaction conclusion is stable or link-conditional. This shows which part of the conclusion depends on modeling conventions and which part does not. Stable conclusions across plausible specifications are easier to defend. Unstable conclusions should be reported as conditional on the modeling choice, as choices that affect the claim should be visible, justified, and interpretable (Mustillo et al., 2018).

Table 4

Practical starting points for link-aware interaction testing across common psychological outcomes.

Outcome type	Plausible starting point	Minimal sensitivity check
Forced-choice accuracy	Binomial logit/probit; chance-corrected link when the task floor is part of the estimand.	Compare standard logit/probit and chance-corrected links; report convergence and response-scale contrasts.
Binary/proportion outcome	Binomial logit/probit; consider cloglog for asymmetric processes.	Compare logit/probit; inspect calibration and contrasts near bounds or rare-event regions.
Likert/ordinal item	Ordinal logit/probit; metric model only with a clear observed-scale target.	Compare ordinal and metric summaries; report category probabilities or response-scale contrasts.
Sum score/composite	Gaussian identity when many levels stay far from bounds; otherwise consider item-level, ordinal, or bounded-score models.	Inspect bounds, skew, and score concentration; compare with item-level or bounded-score sensitivity models when feasible.
Counts/error counts	Poisson or negative binomial, usually with log link; identity only with an observed-count target.	Check overdispersion and zero inflation; compare count contrasts across plausible count models.
Response time/skewed positive outcome	Lognormal, Gamma log/inverse, Inverse Gaussian, or transformed Gaussian, matched to absolute, proportional, reciprocal, or transformed change.	Compare raw-, log-, and response-scale summaries; inspect tail fit and predicted values.

Discussion

Whether an interaction claim survives plausible link alternatives is a question that existing reporting practice rarely answers. Some claims are stable across defensible links and response-scale summaries; others hold only under a modeling choice that is currently left implicit. The scale-dependence of interactions is well known (Cox, 1984; Loftus, 1978; Wagenmakers et al., 2012), and in GLMs it takes the concrete form developed above, where the link fixes the scale on which effects are additive.

The preregistered review showed why this matters for current practice. Interaction tests were common, link functions were often implicit, and many interaction-testing articles included identity-link analyses with outcomes for which observed-scale additivity was not self-evident. The three simulation studies showed the risk of family and link misspecifications. The simulations were designed to quantify pseudo-interactions under known conditions, not to estimate how often they arise in the literature. Forced-choice accuracy is the clearest case, where the binomial family is correct but a standard link misplaces the lower asymptote. The logit-probit comparison shows that the problem persists even within a family, between links that look similar in the central range. Sum scores extend it to the target metric, where the manifest score is already a measurement mapping from item responses.

These results complement work showing that interaction effects are often difficult to estimate precisely even when the model form is correct (Baranger et al., 2023; Castillo et al., 2026). They show how link and metric choices can produce statistically significant product terms even when the true interaction is zero on the generating scale. The diagnostic simulation adds a practical caution. Residual checks, Pregibon-style tests, and information criteria can be informative, but their diagnostic value was uneven and depended on the candidate set. The chance-floor scenario is the clearest relevant failure case, where a wrong link produced a large pseudo-interaction while residual checks showed no sign of misfit and AIC favored the simpler standard link. This is diagnostic misalignment, not uniform diagnostic failure. Diagnostics should therefore be treated as evidence about model fit within a candidate set, not as proof that an interaction claim is robust to link choice (Dunn & Smyth, 1996; Hartig, 2024; Pregibon, 1980).

The concern reaches well outside GLMs. A closely related pattern appears in regression mixture models, where the assumption of normally distributed errors within latent classes can make distributional misspecification look like population heterogeneity (Van Horn et al., 2012). When within-class normality is violated, even modest departures may produce extra classes that reflect skewness, boundedness, scale compression, or other unmodeled distributional features. These are pseudo-clusters, in the same descriptive sense in which the product terms examined here are pseudo-interactions (Toffalini et al., 2024). In both cases, a restrictive metric or distributional assumption turns continuous nonlinear structure into apparent substantive structure, extra classes in one setting and product-term interactions in ours. Artifacts of this kind, arising where the assumed scale meets a boundary, are a general modeling risk rather than a feature of interaction testing alone.

Four limitations bound the scope of the conclusions. First, the review was descriptive and limited to five reference journals across major areas of psychology. Its numerical estimates should therefore be read as reflecting reporting practice in high-standard outlets during the sampled year. A broader mapping across journals, years, and subfields could later support reporting guidance on interaction tests with constrained outcomes. Second, the review coded directly observable reporting features rather than judging, for each article, whether its interaction was scale-dependent enough to be removable. Third, the simulations are mechanism demonstrations aimed at a specific regime, two established main effects that jointly expose observations to link-function curvature. They isolate scale problems under known data-generating processes and do not estimate prevalence or cover all realistic complications. Finally, logit and probit often lead to very similar conclusions, especially away from boundaries, and for prediction, the link is largely a computational convenience, and what matters is forecast accuracy and staying within admissible bounds, such as probabilities between 0 and 1; our claim is that link choice can matter for interaction claims, not that it always does.

Interaction effects are routinely reported as though the modeling scale were a neutral backdrop rather than a substantive choice. Link-aware testing corrects this by making three usually-conflated judgments explicit: which outcome family is plausible, which link defines the scale on which additivity holds, and whether the conclusion survives plausible alternatives to that choice. This reframing does more than check robustness, as it tells researchers what kind of claim they can defend. Stability across links licenses a claim about the interaction itself; instability licenses a narrower claim about the interaction on a particular scale, which is still a defensible finding if reported as such rather than discarded as a failed robustness check.

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Appendix

Supplementary

The simulation figures in the main text summarize the false-positive results. The tables below give the per-scenario numbers behind those figures.

Table A1

Simulation 1: false-positive interaction rates and median model-implied observed-scale changes in the group gap for forced-choice accuracy data, by performance scenario. Fits is the number of the replications in which the model converged; Sig is the number returning a significant age-by-group product term. CI = Wilson 95% confidence interval.

Scenario	Model	Fits	Sig	False-positive rate	95% CI	Median gap change
Lower performance	Gaussian identity	300	137	0.457	[0.401, 0.513]	-1.779
Lower performance	Standard binomial logit	300	191	0.637	[0.581, 0.689]	-1.769
Lower performance	Standard binomial probit	300	183	0.610	[0.554, 0.663]	-1.781
Lower performance	Chance-corrected binomial logit	179	13	0.073	[0.043, 0.120]	-1.233
Middle performance	Gaussian identity	300	40	0.133	[0.099, 0.176]	-0.865
Middle performance	Standard binomial logit	300	169	0.563	[0.507, 0.618]	-0.862
Middle performance	Standard binomial probit	300	148	0.493	[0.437, 0.550]	-0.866
Middle performance	Chance-corrected binomial logit	300	18	0.060	[0.038, 0.093]	-0.707
Higher performance	Gaussian identity	300	36	0.120	[0.088, 0.162]	0.654
Higher performance	Standard binomial logit	300	103	0.343	[0.292, 0.399]	0.555
Higher performance	Standard binomial probit	300	59	0.197	[0.156, 0.245]	0.617
Higher performance	Chance-corrected binomial logit	300	14	0.047	[0.028, 0.077]	0.526

Note. Successful fits are the number of replications in which the model converged and returned an interaction test. Estimates for models with failed fits are summarized over successful replications only.

Table A2

Simulation 2: false-positive interaction rates for the condition-by-group product term in repeated-trial random-intercept GLMMs. CI = Wilson 95% confidence interval.

Generating link	Fitted link	Link status	Fits	Sig	False-positive rate	95% CI	Median beta interaction	Median SE
logit	logit	Matched link	300	16	0.053	[0.033, 0.085]	-0.001	0.075
logit	probit	Wrong link	300	45	0.150	[0.114, 0.195]	-0.041	0.042
probit	logit	Wrong link	300	41	0.137	[0.102, 0.180]	0.071	0.076
probit	probit	Matched link	300	15	0.050	[0.031, 0.081]	-0.003	0.043

Table A3

Simulation 3: false-positive interaction rates and median model-implied observed-scale changes in the group gap for sum-score data. Data were generated with no x-by-group product term on the latent scale. The latent generating scale is an idealized benchmark available only because theta is known in the simulation. CI = Wilson 95% confidence interval.

Scenario	Model	Scale	Fits	Sig	False-positive rate	95% CI	Median gap change
Middle of scale	Gaussian-probit bounded score	sum score	300	14	0.047	[0.028, 0.077]	-0.568
Middle of scale	Latent generating scale	latent theta	300	16	0.053	[0.033, 0.085]	0.018
Middle of scale	Observed sum-score identity	sum score	300	29	0.097	[0.068, 0.135]	-0.547
Near ceiling	Gaussian-probit bounded score	sum score	300	15	0.050	[0.031, 0.081]	1.413
Near ceiling	Latent generating scale	latent theta	300	16	0.053	[0.033, 0.085]	0.003
Near ceiling	Observed sum-score identity	sum score	300	112	0.373	[0.321, 0.429]	1.328
Near floor	Gaussian-probit bounded score	sum score	300	26	0.087	[0.060, 0.124]	-1.847
Near floor	Latent generating scale	latent theta	300	18	0.060	[0.038, 0.093]	0.011
Near floor	Observed sum-score identity	sum score	300	216	0.720	[0.667, 0.768]	-1.762